

Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 2

Winter Semester 2024-2025

DURATION: 30 MIN

FULL MARKS: 20

CSE 4107: Structured Programming I

Figures in the right margin indicate marks.

1. Find the output of the following program:

```
#include <stdio.h>
int main(){
    int i,j;
    for(i=0,j=10;i<j;i++,j--)
    {
        if(j%2==0){
            printf("%d %d\n",i,j);
        }
    }

    return 0;
}
```

6
(CO1)
(PO2)

2. Write a C program that takes a positive integer, n , from the user and prints a triangle using the '*' character, where the size of the triangle is determined by n . A sample input and output pair is shown below:

```
Enter n: 5
*
***
*****
*****
*****
```

8
(CO3)
(PO1)

3. Explain the differences in the execution of the statements between the following two code snippets:

```
if (condition)
    statement1;
else
    statement2;
```

Code Snippet 1

```
if (condition) {
```

6
(CO2)
(PO1)

```
        statement1;  
    }  
    statement2;
```

Code Snippet 2